Probability and Statistic (for CS Students)

Lecture 11: Other Discrete Distributions

University of Mazandaran

Fall 1405

Recap: Bernoulli and Binomial Random Variables

The number of successes, in n independent trials, where each trial is a Bernoulli experiment with success probability p:

$$X \sim \text{Bin}(n, p)$$

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$$

Recap: Bernoulli and Binomial Random Variables

Binomial Random Variable

Notation: $X \sim \text{Bin}(n, p)$

Description: Number of "successes" in n identical, independent experiments each with probability of success p.

Parameters: $n \in \{0, 1, \ldots\}$, the number of experiments.
$p \in [0, 1]$, the probability that a single experiment gives a "success".

Support: $x \in \{0, 1, \ldots, n\}$

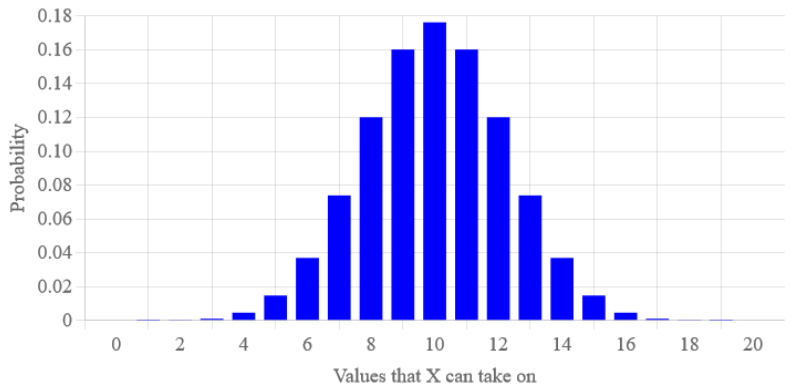
PMF equation: $\mathrm{P}(X = x) = \binom{n}{x} p^x (1-p)^{n-x}$

Expectation: $\mathrm{E}[X] = n \cdot p$

Variance: $\mathrm{Var}(X) = n \cdot p \cdot (1-p)$

PMF graph:

Parameter n: 20 Parameter p: 0.5



Bernoulli Random Variable

Notation: $X \sim \text{Bern}(p)$

Description: A boolean variable that is 1 with probability p

Parameters: p, the probability that $X = 1$.

Support: x is either 0 or 1

PMF equation: $\mathrm{P}(X = x) = \begin{cases} p & \text{if } x = 1 \\ 1-p & \text{if } x = 0 \end{cases}$

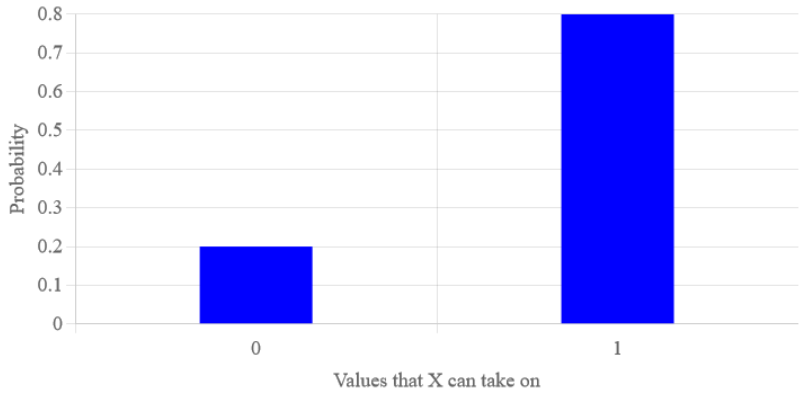
PMF (smooth): $\mathrm{P}(X = x) = p^x (1-p)^{1-x}$

Expectation: $\mathrm{E}[X] = p$

Variance: $\mathrm{Var}(X) = p(1-p)$

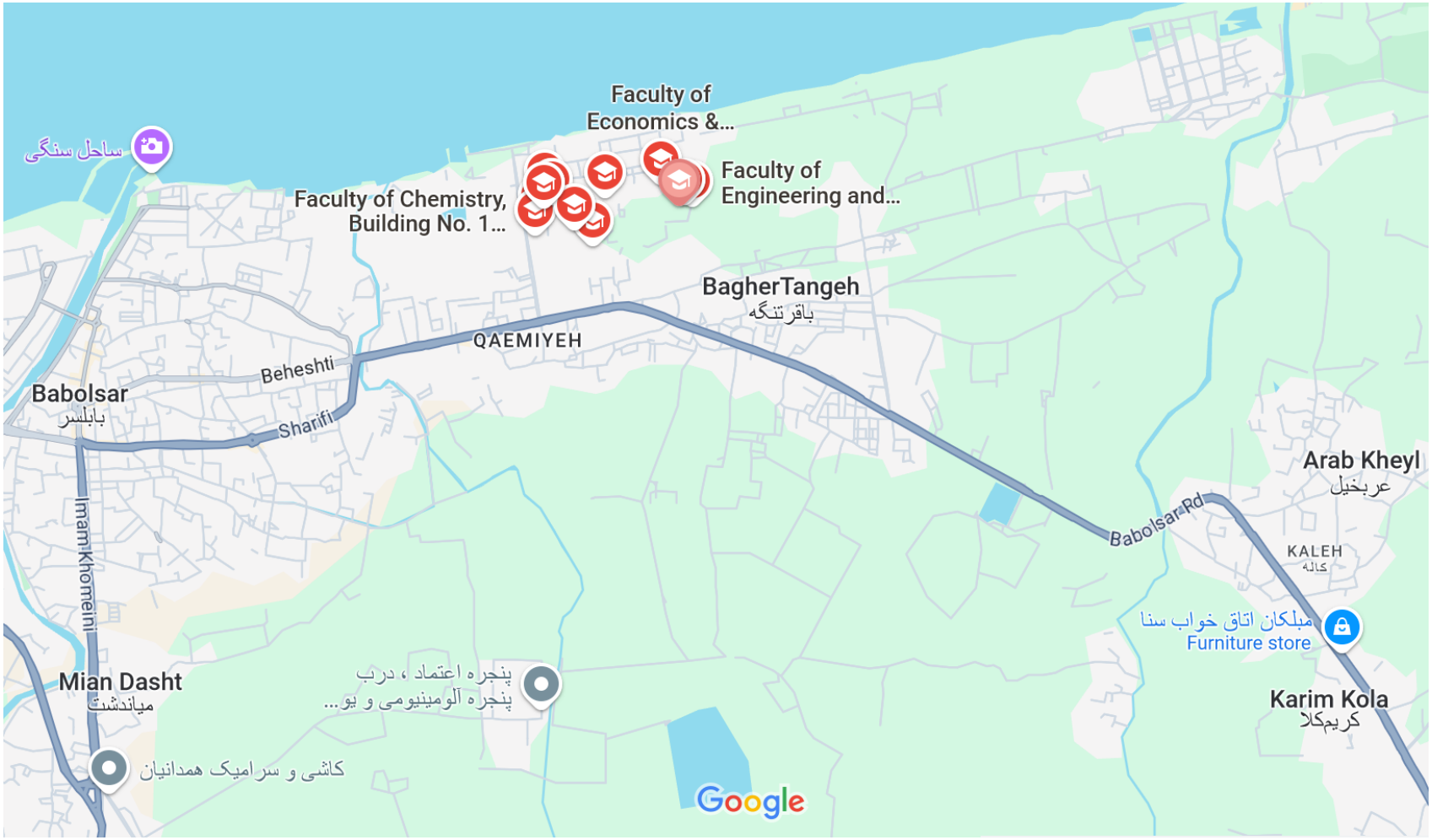
PMF graph:

Parameter p: 0.80



Recap: Poisson Random Variable

Probability of k ride request from this area in the next 1 min



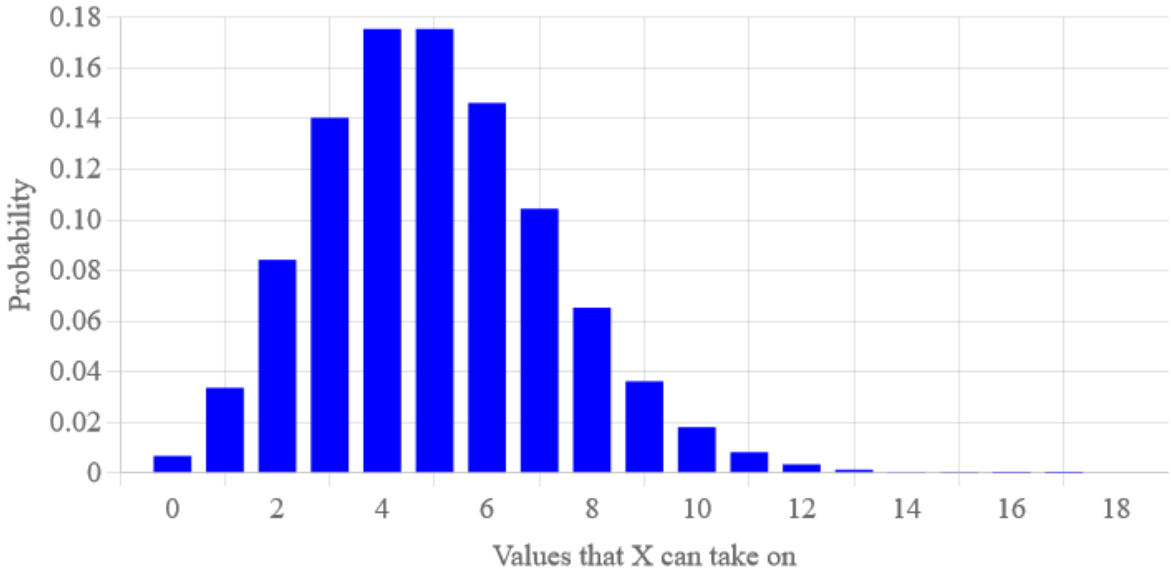
Recap: Poisson Random Variable

Poisson Random Variable

Notation:	$X \sim \text{Poi}(\lambda)$
Description:	Number of events in a fixed time frame if (a) the events occur with a constant mean rate and (b) they occur independently of time since last event.
Parameters:	$\lambda \in \mathbb{R}^+$, the constant average rate.
Support:	$x \in \{0, 1, \ldots\}$
PMF equation:	$\text{P}(X = x) = \frac{\lambda^x e^{-\lambda}}{x!}$
Expectation:	$\text{E}[X] = \lambda$
Variance:	$\text{Var}(X) = \lambda$

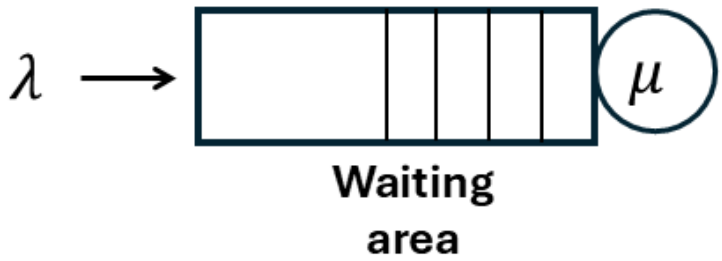
PMF graph:

Parameter λ: 5



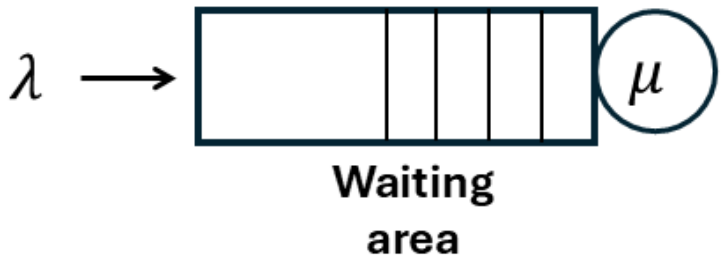
Servers and Their Request Waiting Time

A server wants to know how long a request should wait in the waiting area before it gets served.



Servers and Their Request Waiting Time

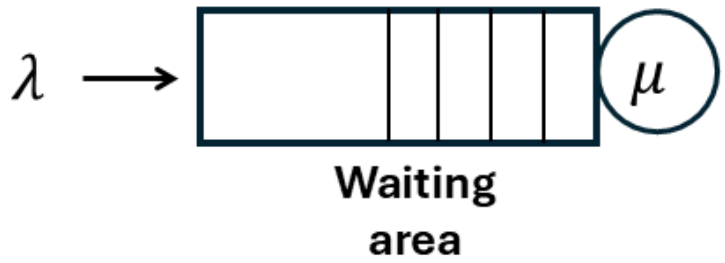
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One of the most common algorithms is MD1 queue

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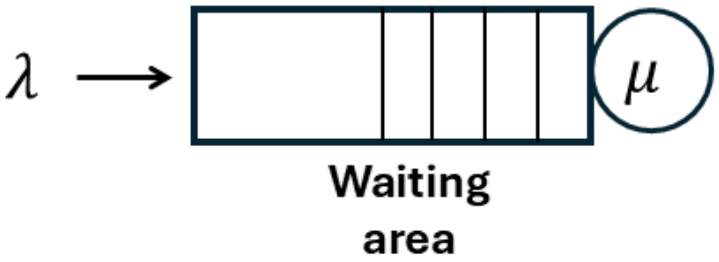


One of the most common algorithms is MD1 queue

You find a paper that says the length of a server "busy period" is distributed as a Borel with parameter $\mu = 0.2$

Servers and Their Request Waiting Time

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Borel distribution

3 languages

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From Wikipedia, the free encyclopedia

The **Borel distribution** is a discrete probability distribution, arising in contexts including branching processes and queueing theory. It is named after the French mathematician Émile Borel.

If the number of offspring that an organism has is Poisson-distributed, and if the average number of offspring of each organism is no bigger than 1, then the descendants of each individual will ultimately become extinct. The number of descendants that an individual ultimately has in that situation is a random variable distributed according to a Borel distribution.

Borel distribution

Parameters	$\mu \in [0,1]$
Support	$n \in \{1,2,3,\ldots\}$
PMF	$\frac{e^{-\mu n}(\mu n)^{n-1}}{n!}$
Mean	$\frac{1}{1-\mu}$
Variance	$\frac{\mu}{(1-\mu)^3}$

Definition [edit]

A discrete random variable X is said to have a Borel distribution[1][2] with parameter $\mu \in [0,1]$ if the probability mass function of X is given by

$$P_\mu(n) = \Pr(X = n) = \frac{e^{-\mu n}(\mu n)^{n-1}}{n!}$$

for $n = 1, 2, 3 \ldots$.

Some More Random Variables

	number of successes	time to get successes	
One Trial	$X \sim \text{Ber}(p)$	$X \sim \text{Geo}(p)$	One Success
	$\uparrow n = 1$	$\uparrow r = 1$	
Several Trials	$X \sim \text{Bin}(n, p)$	$X \sim \text{NegBin}(r, p)$	Several Successes
Interval of Time	$X \sim \text{Poi}(\lambda)$	$X \sim \text{Exp}(\lambda)$	One Success

Geometric Random Variable

X is Geometric Random Variable: $X \sim \text{Geo}(p)$

- X is the number of independent trials until first success
- p is probability of success on each trial
- X takes on values $1, 2, 3, \cdots$ with probability:

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- X takes on values $1, 2, 3, \cdots$ with probability:

$$P(X = n) = (1-p)^{n-1} p$$

$$E[X] = \frac{1}{p}$$

$$\text{Var}(X) = \frac{1-p}{p^2}$$

Negative Binomial Random Variable

X is Negative Binomial Random Variable: $X \sim \text{NegBin}(r, p)$

- X is number of independent trials until r successes
- p is probability of success on each trial
- X takes on values $r, r+1, r+2, \cdots$ with probability

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$$P(X = n) = \binom{n-1}{r-1} p^r (1-p)^{n-r}, \text{where } n = r, r+1, \cdots$$

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- X is number of independent trials until r successes
- p is probability of success on each trial
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$$P(X = n) = \binom{n-1}{r-1} p^r (1-p)^{n-r}, \text{where } n = r, r+1, \cdots$$

$$E[X] = \frac{r}{p}$$

$$\text{Var}(X) = r\frac{1-p}{p^2}$$

Geometric and Negative Binomial

Geometric Random Variable

Notation: $X \sim \text{Geo}(p)$

Description: Number of experiments until a success. Assumes independent experiments each with probability of success p.

Parameters: $p \in [0, 1]$, the probability that a single experiment gives a "success".

Support: $x \in \{1, \ldots, \infty\}$

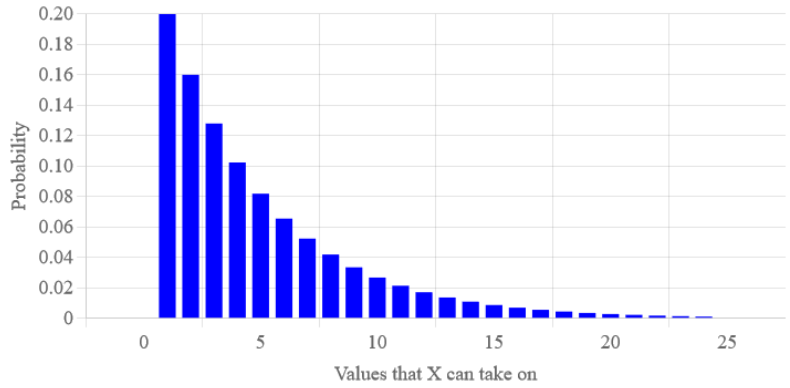
PMF equation: $\text{P}(X = x) = (1-p)^{x-1}p$

Expectation: $\text{E}[X] = \frac{1}{p}$

Variance: $\text{Var}(X) = \frac{1-p}{p^2}$

PMF graph:

Parameter p: 0.20



Negative Binomial Random Variable

Notation: $X \sim \text{NegBin}(r, p)$

Description: Number of experiments until r successes. Assumes each experiment is independent with probability of success p.

Parameters: $r > 0$, the number of success we are waiting for.
$p \in [0, 1]$, the probability that a single experiment gives a "success".

Support: $x \in \{r, \ldots, \infty\}$

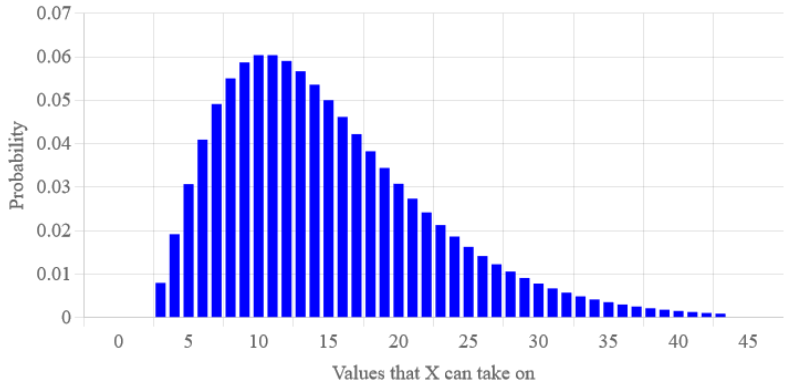
PMF equation: $\text{P}(X = x) = \binom{x-1}{r-1} p^r (1-p)^{x-r}$

Expectation: $\text{E}[X] = \frac{r}{p}$

Variance: $\text{Var}(X) = \frac{r \cdot (1-p)}{p^2}$

PMF graph:

Parameter r: 3 Parameter p: 0.20



Overview of Discrete Distributions

Bernoulli

- indicator of coin flip $X \sim \text{Ber}(p)$

Binomial

- # successes in n coin flips $X \sim \text{Bin}(n, p)$

Poisson

- # successes in a fixed interval of time $X \sim \text{Poi}(\lambda)$

Geometric

- # coin flips until success $X \sim \text{Geo}(p)$

Negative Binomial

- # trials until r successes $X \sim \text{NegBin}(r, p)$

The Challenge of Equity

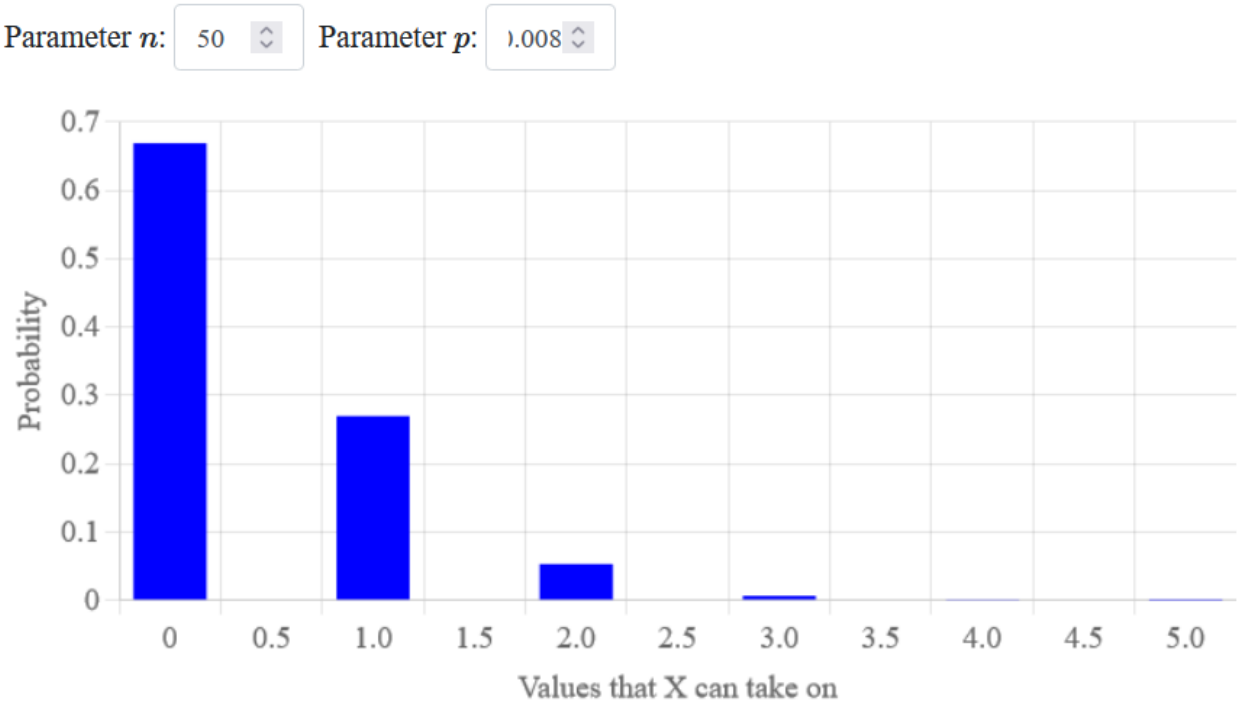
Say, we want to choose a group of 50 students at UMZ and ask them about the most important issues in UMZ to be address asap.

Assume out of 12500 students at UMZ, only 100 are international students. Are we biased against non-Iranian students?

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Bitcoin Mining



SHA-256 Hash(*Data Fixed*, Salt Choice)

Number that looks like random bits

You "mine a bitcoin" if, for given data D, you find a salt number N such that Hash(D, N) produces a string that starts with g zeroes.

Bitcoin Mining

(a) What is the probability that the first number you try will produce a bit string which starts with g zeros (in other words, you mine a bitcoin)

(b) What is the probability that you will need under 100 attempts to mine 2 bit coins?

Bitcoin Mining

(a) What is the probability that the first number you try will produce a bit string which starts with g zeros (in other words, you mine a bitcoin)

Let X be the number of zeros in the first g bits.

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Bitcoin Mining

(a) What is the probability that the first number you try will produce a bit string which starts with g zeros (in other words, you mine a bitcoin)

Let X be the number of zeros in the first g bits.

$X \sim \text{Bin}(n = g, p = 0.5)$

(b) What is the probability that you will need under 100 attempts to mine 2 bit coins?

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Let X be the number of zeros in the first g bits.

$X \sim \text{Bin}(n = g, p = 0.5)$

$P(X = g) = \left(\frac{1}{2}\right)^g$

(b) What is the probability that you will need under 100 attempts to mine 2 bit coins?

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$P(X = g) = \left(\frac{1}{2}\right)^g$ call this p_a

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Let X be

- # bitcoins mined in the first 99 attempts

  $X \sim \text{Bin}(n = 99, p = p_a)$

Bitcoin Mining

(a) What is the probability that the first number you try will produce a bit string which starts with g zeros (in other words, you mine a bitcoin)

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$P(X = g) = \left(\frac{1}{2}\right)^g$ call this p_a

(b) What is the probability that you will need under 100 attempts to mine 2 bit coins?

Let X be

- # bitcoins mined in the first 99 attempts
  $X \sim \text{Bin}(n = 99, p = p_a)$
- # attempts until 2 bitcoins mined
  $X \sim \text{NegBin}(r = 2, p = p_a)$

Bitcoin Mining

(a) What is the probability that the first number you try will produce a bit string which starts with g zeros (in other words, you mine a bitcoin)

Let X be the number of zeros in the first g bits.

$X \sim \text{Bin}(n = g, p = 0.5)$

$P(X = g) = \left(\frac{1}{2}\right)^g$ call this p_a

(b) What is the probability that you will need under 100 attempts to mine 2 bit coins?

Let X be

- # bitcoins mined in the first 99 attempts
  $X \sim \text{Bin}(n = 99, p = p_a)$
- # attempts until 2 bitcoins mined
  $X \sim \text{NegBin}(r = 2, p = p_a)$

$$P(X < 100) = \sum_{x=2}^{99} P(X = x)$$